

Fig. 2: Leakage points where devices can leak out of the system and get into the market

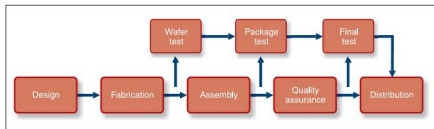


Fig. 3: Devices that are rejected in one of the test stages in the manufacturing of an IC

from factories. By forging documentation, devices can be up-marked (represented as a higher specification part) and sold at a higher price. A good example is to mark a commercial-grade part as industrial-grade.

Defects found in counterfeits

There are two types of defects: internal or invisible defects and external or visible defects. Internal defects are generally called package defects, whereas external defects are further classified into two categories:

Procedural defects. These mainly relate to the packaging and shipping of components and their markings

Mechanical defects. These are due to structural deficiencies and can be further classified as:

Leads/balls/columns. Damages found in leads of different IC packages

Package dimensions and type. Deviations in the IC package from standard packages as defined by JEDEC standards

External defects occur due to reuse of devices, processes used in getting the devices ready for reuse, especially while pulling out of PCBs.

Internal defects are not visible and are invariably associated with the internals of ICs, which could have happened either in the foundry or at the package-assembly stage.

When ICs are manufactured, the die is attached to wire frames. Depending on the design parameters, designers use either a single wire or two wires for bonding the die to the leads. Most counterfeit ICs have either one or both these burnt due to usage.

Another internal fault is the damaged die inside ICs. This happens either due to the process or delamination. At this stage, we need to remember that a counterfeit may not be functional.

Detecting counterfeits

Detection of counterfeits is a time-consuming and intensive process. Proper supply chain checks need to be in place for detecting counterfeits early on in the process; detecting these just as these enter the inventory is the best way to avoid problems.

There are several tests that could be performed to detect counterfeits.

First is a physical test, using incoming inspection or an automated image-recognition system for inspecting the information printed on the package.

Second is a destructive test in which samples are physically destroyed to find counterfeits.

The third uses sophisticated tests like X-ray spectrometry or material analysis for accurate detection.

Another type of detection involves electrical parameter testing. These tests either check the electrical parameters or subject the counterfeits to burn-ins to check durability of parts. At times, all these tests are carried out to identify counterfeits.

How to avoid counterfeiting

Avoiding counterfeits is a tricky and expensive process. However, compared to the cost of the bad impact of counterfeits on products, a little price paid for avoiding is better in the long run. Avoiding counterfeit parts needs proactive and real-time actions.

First step is to control the supply chain so that the purchase process is robust, and all data of purchased components is logged and kept for future reference. This data is typically captured and kept when avionics and medical devices are manufactured. For other products, it is basically the manufacturing process that addresses this aspect.

Proactive avoidance mechanism in the design and manufacturing of ICs makes counterfeiting as difficult as possible. Proactive avoidance techniques include avoiding die and IC recycling (includes two methods of combating counterfeiting, namely, anti-fuse based avoidance and ring oscillator based avoidance), watermarking of ICs, physical unclonable functions and secure split tests.

Let us now see how counterfeiting is being tackled at design level.

Combating die and IC recycling (CDIR). Bulk counterfeiting happens at foundry and assembly locations, and there are two basic technologies that are used. First is anti-fuse/fuse based technology, which is similar to the technology used in programmable logic devices.

Essentially, when an IC with anti-fuse protection powers up, for a brief moment, the programmable logic is in read mode and the central processing unit (CPU) is able to read and verify the authenticity of the device by comparing it with the data supplied by the vendor. Since it requires programming